

E-COMMERCE BUSINESS ANALYSIS REPORT.

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Overview

This analysis was conducted to evaluate product performance, customer behavior, payment preferences, promotional effectiveness, and marketing channel performance using sales transaction data from the DecodeLabs dataset. The goal was to uncover actionable business insights that can improve revenue generation, customer retention, and operational efficiency.

Business Questions, Insights, and Recommendations.

1. Identify the best-selling products by total revenue and total number of orders?

Finding:

The **Printer** product category emerged as the best-performing product across all metrics:

- Highest revenue generated: **\$44,397.41**
- Highest units sold: **124 units**
- Highest customer orders: **42 orders**

On the other hand, **Phones** recorded the lowest sales performance.

Recommendation:

- Increase inventory and promotions for Printers since they are the strongest revenue drivers.
- Investigate why Phones are underperforming by reviewing pricing, product quality, competition, or customer demand.
- Bundle slow-moving products with top-performing items to improve sales.

2. Which products are being cancelled or returned the most?

Finding:

Printers, Laptops, and Tablets recorded the highest lost revenue from failed or returned orders, with Printers having the largest number of failed orders.

Recommendation:

- Improve quality assurance and product testing before shipping.
 - Provide clearer product descriptions and setup guides to reduce customer dissatisfaction.
 - Strengthen after-sales support and return management processes.
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3. Which customers have successfully spent more than \$1,000?

Finding:

Several customers spent between **\$1,000 and \$3,353** in a single successful purchase. Customer **C16775** recorded the highest spending.

Recommendation:

- Create a VIP or loyalty program for high-value customers.
- Offer personalized discounts and exclusive offers to encourage repeat purchases.
- Track high-spending customers for targeted marketing campaigns.

4. What are the most popular payment methods for successful orders?

Finding:

The leading payment methods were **Online Payment** and **Gift Card**, each generating over **\$55,000** in revenue and recording the highest transaction volume.

Recommendation:

- Optimize and simplify online payment systems for better customer experience.
- Promote gift card campaigns during festive or promotional periods.
- Introduce incentives for digital payments to encourage faster transactions.

5. Which promotions based on coupon code are actually driving the most revenue?

Finding:

- **SAVE10** generated the highest revenue (**\$62,231.95**).
- **FREESHIP** was the most frequently used coupon.
- **WINTER15** recorded the lowest usage and revenue.

Recommendation:

- Continue investing in high-performing promotions like **SAVE10**.
- Analyze why **WINTER15** underperformed and redesign future seasonal campaigns.
- Combine free shipping offers with spending thresholds to increase order value.

6. Which customers are consistently coming back to shop?

Finding:

The analysis showed that the maximum number of successful orders per customer was only **1 order**, indicating poor customer retention and lack of repeat purchases.

Recommendation:

- Introduce customer retention strategies such as loyalty rewards, follow-up emails, and personalized recommendations.
- Improve customer engagement after purchase.
- Focus not only on customer acquisition but also on long-term customer relationships.

7. Which referral Sources generate the most orders and their overall percentage shares?

Finding:

Instagram was the top-performing referral source:

- Highest total orders: **256**
- Largest order share: **21.58%**
- Highest revenue generated: **\$275,285.45**

Email marketing also performed strongly, while Referral sources contributed the least.

Recommendation:

- Increase marketing investment in Instagram campaigns and social media advertising.
- Continue strengthening email marketing strategies.
- Improve referral programs by offering incentives for customer referrals.

Conclusion

The analysis reveals that Printers and Instagram are major revenue drivers for the business, while customer retention remains a critical weakness. Promotions such as SAVE10 are highly effective, and online payment methods continue to dominate customer preferences. By improving retention strategies, reducing product returns, and investing in high-performing marketing channels, the business can significantly increase long-term profitability and customer satisfaction.